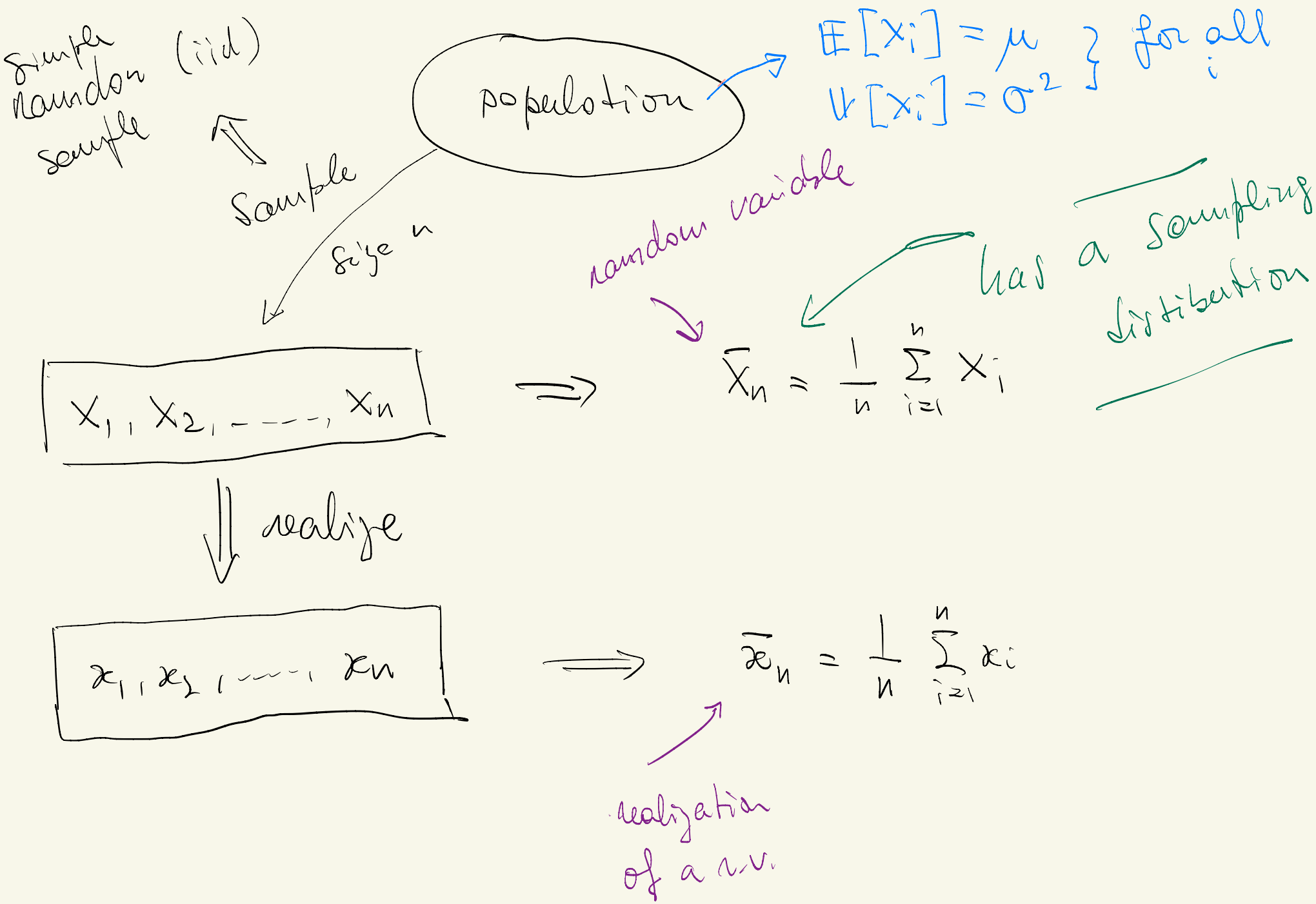
Eco 202 - Fall 2026

Week 13



other statistics : $S_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x}_n)^2$; $X_{(n)} = \max_{1 \le i \le n} X_i$

- Result for all $n = 1, 2, \ldots$

$$\mathbb{E}[\bar{X}_n] = \mu \qquad \mathbb{V}[\bar{X}_n] = \frac{\sigma^2}{n}$$

- Approximations as $n \to \infty$:

  - LLN : $\bar{X}_n \to_p \mu$

  - CLT : $$\frac{\bar{X}_n - \mu}{\sqrt{\mathbb{V}[\bar{X}_n]}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0,1)$$

- A parameter is a non-random feature of the population distribution.

$$\mu = \mathbb{E}[X_i] \quad , \quad \sigma^2 = \mathbb{V}[X_i]$$

- An estimator is a feasible function of the data $(X_1, X_2, \ldots, X_n)$

↳ sampling distributions

Properties of Estimators

- **Unbiasedness**: $\bar{X}_n$ is unbiased for μ if

$$\mathbb{E}[\bar{X}_n] = \mu \quad \text{for all } \mu.$$

$\mathbb{E}[\bar{X}_n] > \mu \Rightarrow$ biased upwards

$\bar{X}_n$ over-estimates μ

$\mathbb{E}[\bar{X}_n] < \mu \Rightarrow$ biased downwards

$\bar{X}_n$ under-estimates μ.

Another example

- Parameter $\sigma^2 = \mathbb{V}[X_i]$
- Estimator $S_n^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(X_i - \bar{X}_n\right)^2$

$$\Rightarrow \mathbb{E}\left[S_n^2\right] = \sigma^2$$

- Alternative: $\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^{n} \left(X_i - \bar{X}_n\right)^2$

$$\Rightarrow \mathbb{E}\left[\hat{\sigma}_n^2\right] = \mathbb{E}\left[\frac{n-1}{n} \cdot S_n^2\right] = \frac{n-1}{n} \cdot \sigma^2$$

Consistency

- $\bar{X}_n$ is consistent for μ if

$$\bar{X}_n \xrightarrow{P} \mu$$

- S_n^2 and $\hat{\sigma}_n^2$ are <u>both</u> consistent for σ^2

because:

$$S_n^2 \xrightarrow{P} \sigma^2$$

$$\hat{\sigma}_n^2 \xrightarrow{P} \sigma^2$$

Mean-squared Error (MSE)

$\hat{\theta}_n$ = estimator $\qquad$ θ = parameter

The MSE of $\hat{\theta}_n$ for θ is:

$$\mathbb{E}\left[(\hat{\theta}_n - \theta)^2\right]$$

Result:

$$\mathbb{E}\left[(\hat{\theta}_n-\theta)^2\right] = \mathbb{E}\left[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n] + \mathbb{E}[\hat{\theta}_n] - \theta)^2\right]$$

$$= \mathbb{E}\left[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])^2 + 2(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])(\mathbb{E}[\hat{\theta}_n] - \theta) + (\mathbb{E}[\hat{\theta}_n] - \theta)^2\right]$$

$$= \mathbb{E}\left[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])^2\right] + 2\,\underbrace{\mathbb{E}\left[(\hat{\theta}_n - \mathbb{E}[\hat{\theta}_n])(\mathbb{E}[\hat{\theta}_n] - \theta)\right]}_{\text{equals zero!}} + \mathbb{E}\left[(\mathbb{E}[\hat{\theta}_n] - \theta)^2\right]$$

$$\mathbb{E}\left[(\hat{\theta}_n - \theta)^2\right] = \mathbb{V}\left[\hat{\theta}_n\right] + \left(\mathbb{E}\left[\hat{\theta}_n\right] - \theta\right)^2$$

Mean squared error of $\hat{\theta}_n$ for θ

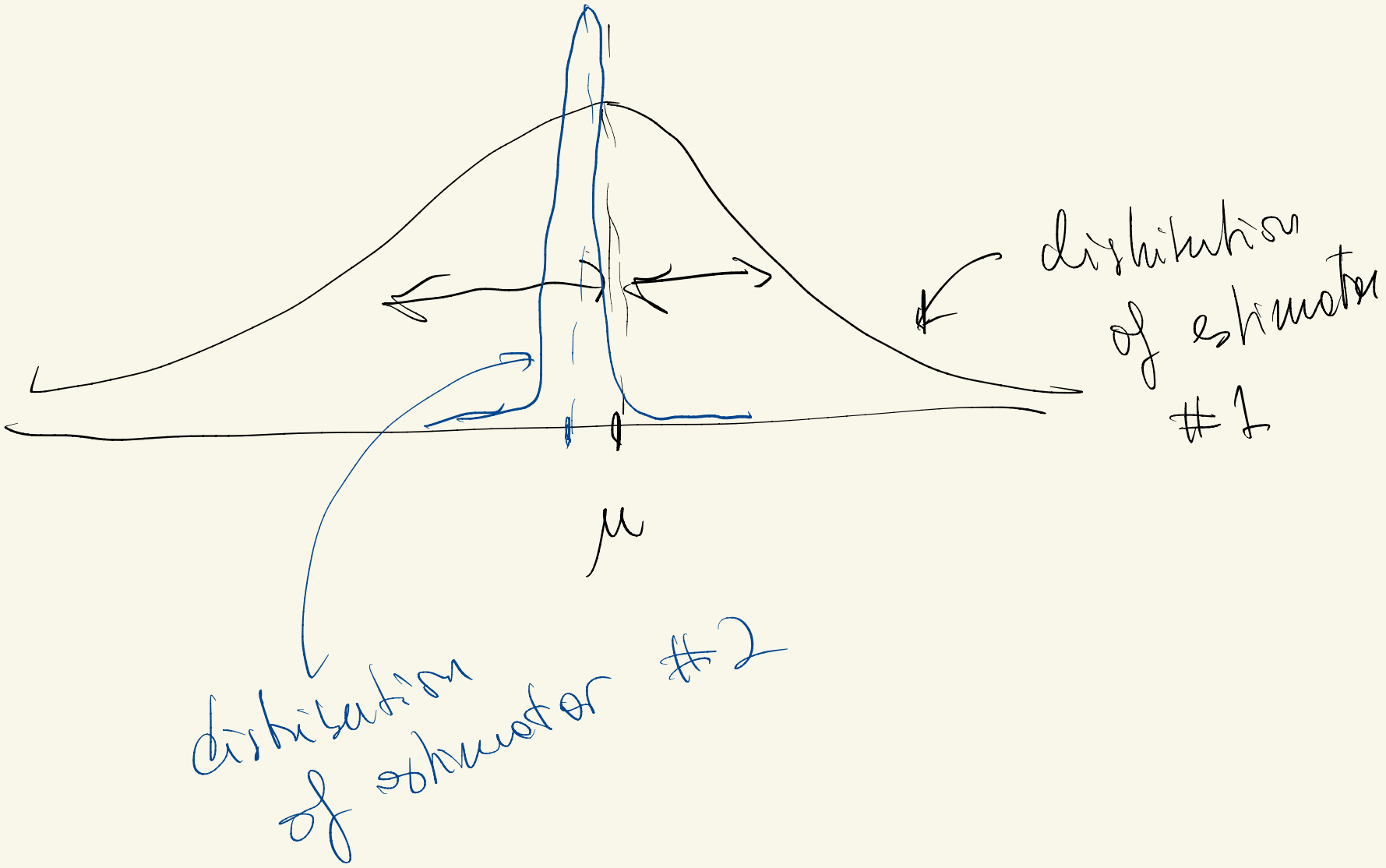
Variance of $\hat{\theta}_n$ + Squared bias of $\hat{\theta}_n$ for θ

Note: the middle term

$$\mathbb{E}\left[\left(\hat{\theta}_n - \mathbb{E}\left[\hat{\theta}_n\right]\right)\left(\mathbb{E}\left[\hat{\theta}_n\right] - \theta\right)\right]$$

$$= \mathbb{E}\left[\left(\hat{\theta}_n - \mathbb{E}\left[\hat{\theta}_n\right]\right)\right] \cdot \left(\mathbb{E}\left[\hat{\theta}\right] - \theta\right)$$

$$= \left(\mathbb{E}\left[\hat{\theta}_n\right] - \mathbb{E}\left[\mathbb{E}\left[\hat{\theta}_n\right]\right]\right) \cdot \left(\mathbb{E}\left[\hat{\theta}\right] - \theta\right) = 0 \quad \square$$


distribution of estimator #1
μ
distribution of estimator #2